

Assignment 4

Optimization for Regression

Recommended completion: Tuesday, October 20, 2026, 11:59 p.m.

Hard deadline: Tuesday, October 27, 2026, 11:59 p.m.

All dates and times are in **Mountain time**. Late submissions are not accepted.

Completion-based grading. A complete submission receives full assignment credit. The notebook's score out of 48 is diagnostic feedback to help you learn; you do not need to pass every test to receive completion credit. Missing submissions receive no credit.

1. Open the template and make your own copy

Open the Assignment 4 student notebook in Google Colab

Sign in with your University of Alberta Google account and select **File** → **Save a copy in Drive**. Work in your own copy so your changes are saved. **The notebook contains additional instructions, explanations, and requirements for each exercise. Read and follow those instructions throughout the notebook to complete the assignment.** This PDF explains how to start and submit. If you use the downloadable notebook from Canvas instead, open it through Colab's **Upload** tab.

2. Run setup and complete the exercises

Run the two setup cells at the top in order. They install the required packages and prepare the data and test cases. **You do not need to download or upload a separate dataset.** The setup needs an internet connection.

ONLY modify code between the following marker lines:

```
### YOUR CODE HERE ###
(replace placeholders here with your solution)
#####
```

Keep the marker lines intact. **Do not modify any provided supporting code outside these blocks:** function names and arguments, initialization, predictor and return code, imports, data loading, examples, plots, setup cells, test cases, or checker calls. Run the provided cells as written. After editing a solution, rerun its cell before checking it.

3. Check your solutions

Each question's quick check, such as `quick_grader.check("q2_1")`, runs two tests while you work. At the end, run `grader.check_all()` to run all 32 tests across the eight questions, including the quick checks. Passing the quick checks alone does not mean all full checks will pass.

The final table shows your diagnostic score and question-level results. Expand a question's feedback to inspect failed tests. You can also run a single question's full checks with `grader.check("q2_1")`.

4. Submit your notebook on Canvas

1. Restart the Colab runtime and run all cells from the beginning. Review any execution errors and the final full-check feedback.
2. Save your notebook, then select **File** → **Download** → **Download .ipynb**.
3. Upload the downloaded `.ipynb` file to **Assignment 4 on Canvas** before the hard deadline. Confirm that Canvas shows your submission.

Submit the notebook file itself, rather than a Colab link, screenshot, or PDF. The instructor uses the same full set of test cases for diagnostic feedback; assignment credit follows the completion policy above.